

## Linear Models Lecture 7: Classic Normal Regression (small sample)

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## Specified Parametric Models

- A model is called **correctly specified** when there exists a unique parameter value  $\theta_0$  such that  $f(x|\theta_0) = f(x)$ , the true data distribution.
- For example, if the true density is

$$f(x) = 2 \exp(-2x)$$

- the exponential model  $f(x|\lambda) = \lambda^{-1} \exp(-x/\lambda)$  is a correctly specified model with  $\lambda_0 = 1/2$
- the lognormal model  $f(x|\mu, \sigma^2) = \frac{1}{x\sigma\sqrt{2\pi}} \exp(-\frac{(\ln x - \mu)^2}{2\sigma^2})$  is misspecified, cannot equal  $2 \exp(-2x)$  under any parameter value.

## Likelihoods

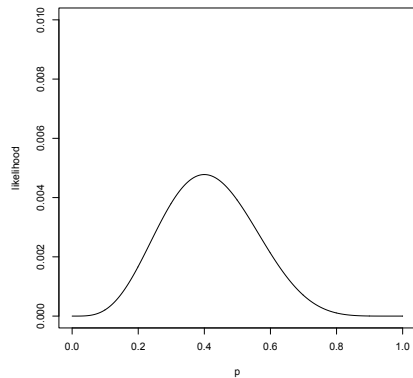
- The density  $f(x|\theta)$  tells us: for a fixed  $\theta$ , how probable is a particular observation  $x$ ?
- The **likelihood** asks the reverse question: for fixed observed data  $\mathbf{X}$ , how plausible is each  $\theta$ ?

$$\mathcal{L}(\theta) \equiv f(X_1, \dots, X_n | \theta) \quad (\text{same formula, viewed as a function of } \theta)$$

- Example: Binomial model of how many terms ( $x$  out of  $n = 2$ ) each of 5 candidates wins.  
 $P(x|p) = \binom{n}{x} p^x (1-p)^{n-x}$ .
- Suppose we observe  $\mathbf{x} = \{1, 0, 1, 2, 0\}$ .
- The Joint Likelihood of the data is:

$$\begin{aligned} P(\mathbf{x} | p) &= \binom{2}{1} p^1 (1-p)^1 \binom{2}{0} p^0 (1-p)^2 \binom{2}{1} p^1 (1-p)^1 \binom{2}{2} p^2 (1-p)^0 \binom{2}{0} p^0 (1-p)^2 \\ &= 4p^4 (1-p)^6 \end{aligned}$$

# Likelihood for Binomial $x$



# Maximum Likelihood Estimator (from Hansen Probability)

## Definition

The **maximum likelihood estimator**  $\hat{\theta}$  of  $\theta$  is the value that maximizes the likelihood:

$$\mathcal{L}_n(\theta) \equiv f(X_1, X_2, \dots, X_n | \theta)$$

Call  $\ell_n = \sum_{i=1}^n \log f(X_i | \theta)$  the log-likelihood.

## Maximizing the likelihood

$$P(\mathbf{x} | p) = 4p^4(1 - p)^6$$

$$\frac{\partial}{\partial p} P(\mathbf{x} | p) = 16p^3(1 - p)^6 - 24p^4(1 - p)^5$$

$$= 0$$

$$3p^4(1 - p)^5 = 2p^3(1 - p)^6$$

$$3p = 2(1 - p)$$

$$5p = 2$$

$$p^* = \frac{2}{5}$$

## Invariance (Useful result)

- Recall that usually,  $E[g(x)] \neq g(E[x])$ , that is, functions of unbiased estimators will not be unbiased.
- If  $\hat{\theta}$  is the MLE of  $\theta$ , then for any transformation  $\beta = h(\theta)$ , the MLE of  $\beta$  is  $\hat{\beta} = h(\hat{\theta})$

Invariance Example  $f(X|\lambda) = \lambda^{-1} \exp(-X/\lambda)$ .

$$\mathcal{L}_n(\lambda) = \prod_{i=1}^n \lambda^{-1} \exp(-X_i/\lambda) = \lambda^{-n} \exp(-\frac{1}{\lambda} \sum_{i=1}^n X_i)$$

$$\log \mathcal{L}_n(\lambda) = -n \log \lambda - \frac{1}{\lambda} \sum_{i=1}^n X_i$$

$$\frac{d\ell_n(\lambda)}{d\lambda} = -\frac{n}{\lambda} + \frac{1}{\lambda^2} \sum_{i=1}^n X_i$$

$$\sum_{i=1}^n X_i = n\bar{X}_n$$

The MLE is  $\hat{\lambda} = \bar{X}_n$

## Invariance Example $\beta = 1/\lambda$

- Set  $\beta = 1/\lambda$ , so  $h(\lambda) = 1/\lambda$ .
- The log density of this model is  $\log f(x|\lambda) = \log[\beta \exp(-x\beta)] = \log \beta - x\beta$ .
- The log likelihood is  $n \log \beta - \beta n \bar{X}_n$
- Take the derivative with respect to  $\beta$ :

$$n/\hat{\beta} - n\bar{X}_n = 0$$

$$\hat{\beta} = 1/\bar{X}_n$$

## From MLE to Precision

We now have an estimator (MLE) and a useful property (invariance). **Two questions remain:**

- 1 How precise is the MLE?** We need a measure of information in the data.
- 2 How precise can any estimator be?** We need a lower bound on variance.

The answers come from three linked concepts:

- **Score:** the derivative of the log-density (the “gradient” that MLE sets to zero).
- **Fisher Information:** the variance of the score (how much the data tells us).
- **Cramér-Rao Bound:** no unbiased estimator can beat  $\text{information}^{-1}$ .

## Score: the slope of the log likelihood

- The **likelihood score** is the derivative of the log-likelihood function:

$$S_n(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta)$$

- The score is a function of  $\theta$  and tells us how sensitive the log-likelihood is to the parameter, and equals zero at the optimum.
- The **efficient score** is the derivative of the log likelihood for a single observation, evaluated at  $\mathbf{x} = X_1, X_2, \dots, X_n$  and the true parameter vector

$$S = \frac{\partial}{\partial \theta} \log f(\mathbf{x}|\theta_0)$$

- The efficient score fixes  $\theta$  at the true value  $\theta_0$ .

## Hessian: the curvature of the log likelihood

- The **likelihood Hessian** is the negative second derivative:

$$\mathcal{H}_n(\theta) = -\frac{\partial^2}{\partial\theta\partial\theta'}\ell_n(\theta)$$

- The Hessian matrix is used to calculate the variance.

## Fisher Information

- The **Fisher information** is the variance of the efficient score (score evaluated at true  $\theta_0$ ).

$$\mathcal{I}_\theta = \mathbb{E}[SS']$$

- The **expected Hessian** is the expectation of the Hessian for a single observation:

$$\mathcal{H}_\theta = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta\partial\theta'} \log f(X|\theta)\right]$$

- If the model is correctly specified:

$$\mathcal{I}_\theta = \mathcal{H}_\theta$$

- **Intuition:** At  $\theta_0$ , the score has mean zero. Both  $\mathcal{I}$  (score variance) and  $\mathcal{H}$  (expected curvature) measure second-order properties of the same density—they agree because these are two ways of measuring the same thing.

## Cramér-Rao Lower Bound

- If  $\tilde{\theta}$  is *any* unbiased estimator of  $\theta$ , then:

$$\text{Var}[\tilde{\theta}] \geq (n\mathcal{I}_{\theta})^{-1}$$

No unbiased estimator can have variance below the inverse of the total Fisher information.

- **Intuition:** More informative data (higher  $\mathcal{I}_{\theta}$ )  $\rightarrow$  tighter bound  $\rightarrow$  more precise estimation is possible. The bound is an information-theoretic limit on what the data can tell you.
- An estimator that achieves the bound is called **efficient**.

## Cramér-Rao vs. Gauss-Markov

- **Gauss-Markov** (Lecture 5): OLS is the best *linear* unbiased estimator, under homoskedasticity. A result specific to the linear model.
- **Cramér-Rao**: No unbiased estimator—linear or not—can beat the information bound. A general result for *any parametric model*.
- Under normality, OLS achieves the CR bound → efficient among **all** unbiased estimators, not just linear ones.
- **The broader point**: The CR bound is one instance of a *semiparametric efficiency bound*. In later lectures, we will see that the same logic—scores, information, influence functions—defines efficient estimators even without a full parametric model.

## Variance Estimators

- The sample Hessian Estimator depends on calculating the second derivatives of the log-likelihood:

$$\hat{\mathcal{H}}_{\theta} = \frac{1}{n} \sum_{i=1}^n -\frac{\partial}{\partial \theta \partial \theta'} \log f(X_i | \hat{\theta})$$
$$\hat{\mathbf{V}}_1 = \hat{\mathcal{H}}_{\theta}^{-1}$$

- The Outer Product Estimator is based on the Fisher Information:

$$\hat{\mathcal{G}}_{\theta} = \frac{1}{n} \sum_{i=1}^n \left( \frac{\partial}{\partial \theta} \log f(X_i | \hat{\theta}) \right) \left( \frac{\partial}{\partial \theta} \log f(X_i | \hat{\theta}) \right)'$$
$$\hat{\mathbf{V}}_2 = \hat{\mathcal{G}}_{\theta}^{-1}$$

## Example: Normal with known mean, unknown variance

$X \sim N(0, \theta)$ , where  $\theta \equiv \sigma^2$ .  $\mathbb{E}[X] = 0$ ,  $\mathbb{E}[X^2] = \theta_0$ ,  $\mathbb{E}[X^4] = 3\theta_0^2$ .

The density is:

$$f(x|\theta) = \frac{1}{(2\pi\theta)^{1/2}} \exp\left(-\frac{x^2}{2\theta}\right)$$

The log density is:

$$\log f(x|\theta) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\theta) - \frac{x^2}{2\theta}$$

the first and second derivatives are:

$$\frac{d}{d\theta} \log f(x|\theta) = -\frac{1}{2\theta} + \frac{x^2}{2\theta^2} = \frac{x^2 - \theta}{2\theta^2}$$

$$\frac{d^2}{d\theta^2} \log f(x|\theta) = \frac{1}{2\theta^2} - \frac{x^2}{\theta^3}$$

## Example: Normal log likelihood

$$\ell_n(\theta) = \sum_{i=1}^n \log f(X_i|\theta) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\theta) - \frac{1}{2\theta} \sum_{i=1}^n X_i^2$$

$$\frac{d}{d\theta} \ell_n(\theta) = -\frac{n}{2\theta} + \frac{1}{2\theta^2} \sum_{i=1}^n X_i^2$$

The MLE is  $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i^2$

## Example: Score and Fisher Information

$$\frac{d}{d\theta} \log f(x|\theta) = \frac{x^2 - \theta}{2\theta^2}$$

The efficient score  $S = \frac{x^2 - \theta_0}{2\theta_0^2}$

$$\text{var}[S] = \frac{\mathbb{E}[(X^2 - \theta_0)^2]}{4\theta_0^4} = \frac{\mathbb{E}[X^4 - 2X^2\theta_0 + \theta_0^2]}{4\theta_0^4} = \frac{3\theta_0^2 - 2\theta_0^2 + \theta_0^2}{4\theta_0^4} = \frac{1}{2\theta_0^2}$$

## Example: Information Equality $\mathcal{I}_\theta = \mathcal{H}_\theta$

$$\mathcal{H}_\theta = \mathbb{E}\left[-\frac{d^2}{d\theta^2} \log f(X|\theta_0)\right] = \mathbb{E}\left[-\frac{d}{d\theta} \frac{x^2 - \theta_0}{2\theta_0^2}\right] = \mathbb{E}\left[\frac{2x^2 - \theta_0}{2\theta_0^3}\right] = \frac{1}{2\theta_0^2}$$

$$\mathcal{I}_\theta = \text{var}[S] = \frac{1}{2\theta_0^2} = \mathcal{H}_\theta$$

The total Fisher information for  $n$  observations is  $I_n(\theta_0) = n \times \mathcal{I}_\theta = \frac{n}{2\theta_0^2}$

The variance of  $\hat{\theta}$  is  $\frac{1}{I_n(\theta_0)} = 2\theta_0^2/n$

So we can get the plug-in estimator for the standard error as  $\sqrt{2\hat{\theta}^2/n}$ .

## Simulation: Fisher Information Standard Errors

```
for(i in 1:100000){  
  x <- rnorm(100, sd=sqrt(5))  
  thetahat[i] <- sum(x^2)/100  
  diff[i] <- thetahat[i]-5  
  se[i] <- sqrt(2*thetahat[i]^2/100) }  
mean(thetahat)  
5.002307  
sqrt(var(diff))  
0.7102136  
mean(se)  
0.707433
```

## What If the Model Is Wrong? Robust Variance

Under correct specification,  $\mathcal{I}_\theta = \mathcal{H}_\theta$  and either one gives the variance.

**Under misspecification**, they diverge:

- $\mathcal{H}$  (curvature) still captures how the estimator behaves locally.
- $\mathcal{I}$  (score variance) captures the actual variability in the data.
- The correct variance requires *both*:

$$\hat{\mathbf{V}} = \hat{\mathcal{H}}^{-1} \hat{\mathcal{I}} \hat{\mathcal{H}}^{-1}$$

This is the **sandwich formula** — the same structure as the heteroskedasticity-robust variance from Lecture 6, now derived from likelihood theory.

## Summary: Scores, Information, and Efficiency

These four concepts are **not specific to MLE** — they recur throughout estimation theory:

- 1 **Score** → the derivative of the log-density. Setting it to zero gives estimating equations — the same objects that define moment conditions and influence functions.
- 2 **Fisher Information** → the variance of the score; measures how much the data can tell you about  $\theta$ .
- 3 **Efficiency bound** → no unbiased estimator can beat  $(n\mathcal{I}_\theta)^{-1}$ . MLE achieves this when the likelihood is correctly specified.
- 4 **Likelihood** → when a full parametric model is available, the likelihood gives us a concrete score and a concrete bound.

**Next:** apply this to the normal linear model and show that OLS *is* the MLE.



## Log Likelihood of the normal linear model

$$\log \mathcal{L}_n(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \beta)^2 \equiv \ell_n(\beta, \sigma^2)$$

## Classic Normal Regression Model

- The Classic Normal Regression Model consists of the following assumptions:

- 1  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$

- 2  $\mathbf{e}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$

- or equivalently:  $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$

- 3  $\text{rank}(\mathbf{X}) = K$ .

In this model,  $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  is the multivariate normal density function whose pdf is:

$$f(\mathbf{e}) = (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \mathbf{e}'\mathbf{e}\right)$$

## Background on Multivariate Normal

- A multivariate normal's parameters are a vector of means and a variance covariance matrix.
- Linear functions of a multinormal vector  $\mathbf{y}$  are also normal.
- If  $\mathbf{z} = \mathbf{g} + \mathbf{H}\mathbf{y}$ , where  $\mathbf{g}$  and  $\mathbf{H}$  are non-random, and  $\mathbf{H}$  has full row rank, then

$$\mathbf{z} \sim N(\mathbf{g} + \mathbf{H}\boldsymbol{\mu}, \mathbf{H}\boldsymbol{\Sigma}\mathbf{H}')$$

- In the case of MVN, independence is the same as uncorrelated.

## Bivariate Normal CEF

- Given two random variables, they are distributed bivariate normal if

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim N \left( \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix} \right)$$

- Call  $X^* = \frac{X - \mu_X}{\sigma_X}$ ,  $Y^* = \frac{Y - \mu_Y}{\sigma_Y}$

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} \exp \left( -\frac{1}{2} (X^*)^2 \right)$$

$$f_{X,Y}(x,y) = \frac{1}{\sqrt{2\pi}\sigma_X\sqrt{2\pi}\sigma_Y} \sqrt{1-\rho^2} \exp \left( -\frac{1}{2[1-\rho^2]} \left[ (X^*)^2 - 2\rho(X^*)(Y^*) + (Y^*)^2 \right] \right)$$

## Bivariate Normal CEF: Derivation

Dividing the joint by the marginal and completing the square:

$$\begin{aligned} f_{Y|X}(y|x) &\propto \exp\left(-\frac{1}{2(1-\rho^2)}(Y^* - \rho X^*)^2\right) \\ &= \exp\left(-\frac{1}{2}\left(\frac{Y - \mu_Y - \rho\frac{\sigma_Y}{\sigma_X}(X - \mu_X)}{\sigma_Y\sqrt{1-\rho^2}}\right)^2\right) \end{aligned}$$

This is the kernel of a normal density with:

- Conditional mean:  $\mathbb{E}[Y|X = x] = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X) = \mu_Y + \frac{\text{Cov}(Y, X)}{\text{Var}(X)}(x - \mu_X)$
- Conditional variance:  $\sigma_Y^2(1 - \rho^2)$

**Key insight:** Under joint normality, the CEF is **linear** and the regression coefficient is  $\beta = \text{Cov}(Y, X)/\text{Var}(X)$ . This is a classical motivation for linear regression.







## R Example: Log-Likelihood and AIC

```
data(swiss)
mod1 <- lm(Fertility ~ Education, data = swiss)
mod2 <- lm(Fertility ~ Education + Agriculture,
           data = swiss)
mod3 <- lm(Fertility ~ Education + Agriculture
           + Catholic + Infant.Mortality, data=swiss)
# Log-likelihoods (higher = better fit)
sapply(list(mod1, mod2, mod3), logLik)
# -168.3 -166.0 -155.3
# AIC = -2*logLik + 2*k (lower = better)
sapply(list(mod1, mod2, mod3), AIC)
# 342.5 340.0 322.6
```

AIC penalizes complexity: mod3 wins because the likelihood improvement outweighs the  $2k$  penalty.

## Score of the Normal Regression Model (Hansen 5.14)

The likelihood scores are the derivatives of the log-likelihood:

$$\frac{\partial}{\partial \beta} \ell_n(\beta, \sigma^2) = \frac{1}{\sigma^2} \sum_{i=1}^n X_i(Y_i - X_i' \beta) = \frac{1}{\sigma^2} \mathbf{X}' \mathbf{e}$$

$$\frac{\partial}{\partial \sigma^2} \ell_n(\beta, \sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (Y_i - X_i' \beta)^2$$

Setting the score for  $\beta$  to zero:

$$\frac{1}{\sigma^2} \mathbf{X}' \mathbf{e} = 0 \quad \Longleftrightarrow \quad \mathbf{X}' \mathbf{e} = 0 \quad \Longleftrightarrow \quad \mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta) = 0$$

These are exactly the **OLS normal equations**! The MLE for  $\beta$  equals the OLS estimator because the score FOC is proportional to the least squares FOC.



## Information Matrix of Normal Regression (Hansen 5.14)

The Fisher information matrix for the normal regression model is:

$$\mathcal{I} = \text{var} \begin{bmatrix} \frac{\partial}{\partial \beta} \ell(\beta, \sigma^2) \\ \frac{\partial}{\partial \sigma^2} \ell(\beta, \sigma^2) \end{bmatrix} = \begin{pmatrix} \frac{1}{\sigma^2} \mathbf{X}'\mathbf{X} & \mathbf{0} \\ \mathbf{0} & \frac{n}{2\sigma^4} \end{pmatrix}$$

- The matrix is **block diagonal**: estimation of  $\beta$  and  $\sigma^2$  are independent.
- The Cramér-Rao lower bound for  $\beta$  is:

$$\mathcal{I}_{\beta}^{-1} = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$$

which is exactly the variance of  $\hat{\beta}_{OLS}$ .

- OLS **achieves the Cramér-Rao bound** — it is efficient among all unbiased estimators under normality.



## MLE as a Special Case of Moment-Based Estimation

**Maximum likelihood** solves  $k$  score equations in  $k$  unknowns:  $\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f(Y_i|X_i, \theta) = 0$

**GMM** solves  $m$  moment conditions in  $k$  unknowns:  $\frac{1}{n} \sum_{i=1}^n g(Y_i, X_i, \theta) = 0$

- When  $m = k$  and  $g = \text{score}$ : **GMM = MLE**. MLE is just-identified GMM.
- When  $m > k$ : GMM uses an *optimal weighting matrix* to combine them.
- IV example:  $\mathbb{E}[Z_i e_i] = 0$  gives more moment conditions than parameters.

**Key insight:**  $\text{MLE} \subset \text{GMM}$ . Score equations are one choice of moment conditions. When the likelihood is misspecified, the sandwich variance replaces the information-based one.

## Summary: The Score-Based View

| Concept     | Formula                                                                             | Role                              |
|-------------|-------------------------------------------------------------------------------------|-----------------------------------|
| Score       | $S_i = \frac{1}{\sigma^2} X_i e_i$                                                  | FOC $\rightarrow$ OLS normal eqns |
| Information | $\mathcal{I}_\beta = \frac{1}{\sigma^2} \mathbf{X}'\mathbf{X}$                      | Precision of $\hat{\beta}$        |
| CR Bound    | $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$                                              | OLS achieves it                   |
| Sandwich    | $(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' D\mathbf{X} (\mathbf{X}'\mathbf{X})^{-1}$ | Robust to misspec.                |
| GMM link    | Score = moment cond.                                                                | MLE $\subset$ GMM                 |

**Next:** exact finite-sample distributions under normality —  $\hat{\beta}$ ,  $s^2$ ,  $t$ , and  $F$ .



## Proof of normality of $\hat{\beta}$ in CNRM

We can show that  $\hat{\beta} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \mathbf{e})$$

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}$$

$$\hat{\beta} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}$$

$$\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}$$

$$\text{Rank}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = \text{Rank}(\mathbf{X}) = K$$

So  $\hat{\beta} - \beta$  is a full-row-rank linear transformation of the multinormal vector  $\mathbf{e}$

$$\hat{\beta} - \beta \sim N(\mathbf{0}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1})$$

## Distribution of Residuals (Hansen 5.7)

Recall  $\hat{\mathbf{e}} = \mathbf{M}\mathbf{e}$  where  $\mathbf{M} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ .

Since  $\mathbf{e}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n)$  and  $\hat{\mathbf{e}}$  is a linear function of  $\mathbf{e}$ :

$$\hat{\mathbf{e}}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{M})$$

**Key fact:**  $\hat{\beta}$  and  $\hat{\mathbf{e}}$  are **independent** (conditional on  $\mathbf{X}$ ).

*Proof:* Their joint covariance is

$$\text{Cov}(\hat{\beta} - \beta, \hat{\mathbf{e}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \cdot \sigma^2\mathbf{I} \cdot \mathbf{M} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \underbrace{\mathbf{X}'\mathbf{M}}_{=0} = \mathbf{0}$$

Since they are jointly normal and uncorrelated, they are independent.

**Why this matters:** It means  $s^2 = \hat{\mathbf{e}}'\hat{\mathbf{e}}/(n - k)$  is independent of  $\hat{\beta}$ . This is what allows us to form the  $t$ -statistic as a ratio of independent normal and  $\chi^2$  components.

## Distribution of residual sum of squares $s_{\hat{\mathbf{e}}}^2 = \frac{\hat{\mathbf{e}}'\hat{\mathbf{e}}}{(N-K)}$ in CNRM

Thm: Let  $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$ , let  $\mathbf{A}$  be idempotent. Then  $\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi^2(\nu)$  where  $\nu = \text{Rank}(\mathbf{A})$ .

$$\begin{aligned}\mathbf{e}|\mathbf{X} &\sim N(\mathbf{0}, \sigma^2\mathbf{I}) \\ \mathbf{e}/\sigma|\mathbf{X} &\sim N(\mathbf{0}, \mathbf{I})\end{aligned}$$

Recall  $\hat{\mathbf{e}} = \mathbf{M}\mathbf{e}$ , so:

$$\begin{aligned}(N-K)s_{\hat{\mathbf{e}}}^2 &= \hat{\mathbf{e}}'\hat{\mathbf{e}} = \mathbf{e}'\mathbf{M}\mathbf{e} \\ \frac{\mathbf{e}'}{\sigma}\mathbf{M}\frac{\mathbf{e}}{\sigma} &\sim \chi^2(N-K) \quad (\mathbf{M} \text{ is idempotent, rank } N-K)\end{aligned}$$

That is, in a linear regression model,  $\frac{(N-K)s_{\hat{\mathbf{e}}}^2}{\sigma^2} \sim \chi_{n-k}^2$

## R Example: $\chi^2$ Distribution of $s^2$

```
set.seed(1); n <- 50; k <- 3; sigma2 <- 4
X <- cbind(1, matrix(rnorm(n*(k-1)), n, k-1))
beta <- c(2, -1, 0.5)
scaled_s2 <- replicate(10000, {
  y <- X %*% beta + rnorm(n, sd = sqrt(sigma2))
  ehat <- resid(lm(y ~ X - 1))
  sum(ehat^2) / sigma2      # (n-k)*s^2 / sigma^2
})
```

```
# Compare to chi-squared(n-k)
mean(scaled_s2)      # ~47 (= n-k)
var(scaled_s2)       # ~94 (= 2*(n-k))
```

Theory:  $\mathbb{E}[\chi_{47}^2] = 47$ ,  
 $\text{Var}[\chi_{47}^2] = 94$ .

The simulation matches.

## From Known $\sigma$ to Estimated $s$ : Why Heavier Tails

If  $\sigma$  were known, the  $j$ -th coefficient is exactly normal:

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}} \sim N(0, 1)$$

But  $\sigma$  is unknown. Replacing it with  $s$  introduces extra randomness:

- When  $s$  underestimates  $\sigma$ , the ratio is *inflated* — producing occasional large values.
- When  $s$  overestimates, the ratio is compressed.
- Net effect: **heavier tails** than the normal, especially when  $n-k$  is small (because  $s$  is noisy with few degrees of freedom).
- As  $n-k \rightarrow \infty$ ,  $s \rightarrow \sigma$  and  $t_{n-k} \rightarrow N(0, 1)$ .



## From Normal to $t$ : Estimated Variance

In practice, we do not know the true error variance  $\sigma^2$ . The unbiased estimator is:

$$s_{\hat{e}}^2 = \frac{1}{n-k} \sum_{i=1}^n \hat{e}_i^2 = \frac{\hat{\mathbf{e}}' \hat{\mathbf{e}}}{n-k}$$

We plug in this estimate to standardize:

$$T = \frac{\hat{\beta}_j - \beta_j}{\sqrt{s_{\hat{e}}^2 [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}}$$

This is no longer standard normal, but it has a  $t$ -**distribution**.



## Simulation: $t$ -Statistics Match the $t$ -Distribution

```
set.seed(1); n <- 15; k <- 3; sigma2 <- 4
X <- cbind(1, matrix(rnorm(n*(k-1)), n, k-1))
beta <- c(2, -1, 0.5)
t_stats <- replicate(10000, {
  y <- X %*% beta + rnorm(n, sd = sqrt(sigma2))
  summary(lm(y ~ X - 1))$coefficients[2, "t value"]
})
```

```
mean(abs(t_stats) > 2)      # 0.067
2*pt(-2, df = n-k)         # 0.065
2*pnorm(-2)                 # 0.046
```

With  $n-k = 12$ , using  $N(0, 1)$  critical values rejects 4.6% instead of 6.5%.





## Summary: Small-Sample Distributions

Under the classical normal regression model:

- $\hat{\beta}|\mathbf{X} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$  *(linear fn of normal)*
- $\hat{e}|\mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{M}), \quad \hat{\beta} \perp \hat{e}$  *(projection)*
- $\frac{(n-k)s^2}{\sigma^2} \sim \chi_{n-k}^2$  *(idempotent quadratic form)*
- $T = \frac{\hat{\beta}_j - \beta_j}{s(\hat{\beta}_j)} \sim t_{n-k}$  *(normal /  $\sqrt{\chi^2/df}$ )*
- $F = \frac{(\tilde{\sigma}^2 - \hat{\sigma}^2)/q}{\hat{\sigma}^2/(n-k)} \sim F_{q,n-k}$  *(likelihood ratio)*

These **exact** results hold in finite samples. Without normality, we rely on asymptotic approximations.



## Confidence Intervals in Practice

$$Pr(\beta \in \hat{C}) = Pr(-c \leq T(\beta) \leq c)$$

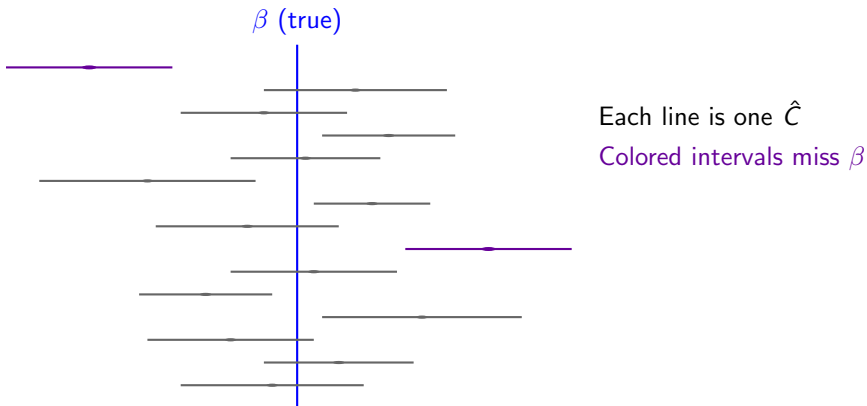
$$Pr(\beta \in \hat{C}) = 2F(c) - 1$$

Our goal is to set this coverage probability equal to  $1 - \alpha$ , or  $F(c) = 1 - \alpha/2$ .

If  $\alpha = .05$ , we solve  $c = F^{-1}(1 - .05/2)$ . In case of a normal,  $c=1.96 \approx 2$

$$\hat{C} = [\hat{\beta} - 2 \cdot s(\hat{\beta}), \hat{\beta} + 2 \cdot s(\hat{\beta})]$$

## Confidence Intervals: The Random Interval Covers a Fixed Parameter



$\beta$  is *fixed*. Each sample yields a different random interval  $\hat{C}$ . About  $1-\alpha$  of them cover  $\beta$ .



## Example: Swiss Fertility Data

|                      | <i>Dep. var: Fertility</i>     |
|----------------------|--------------------------------|
| Education (%)        | −0.98***<br>(0.15)             |
| Agriculture (%)      | −0.15**<br>(0.07)              |
| Catholic (%)         | 0.12***<br>(0.03)              |
| Infant.Mortality (%) | 1.08***<br>(0.38)              |
| Constant             | 62.10***<br>(9.60)             |
| Observations         | 47                             |
| $R^2$                | 0.70                           |
| Residual Std. Error  | 7.17 (df = 42)                 |
| Note:                | * p<0.1; ** p<0.05; *** p<0.01 |

Did education share predict the average fertility across Swiss districts in 1888?

Focus on the Education coefficient:  $\hat{\beta} = -0.98$ ,  $se(\hat{\beta}) = 0.15$ .

The  $t$ -statistic is  $-0.98/0.15 = -6.61$ , with 42 degrees of freedom.

Is this large enough to reject  $\mathbb{H}_0 : \beta = 0$ ?

